

Student Name & ID: Syed Ali Nisar Shah (ss07136) & Syed M. Asad Ali (sa09475)

Date: September 26th, 2025

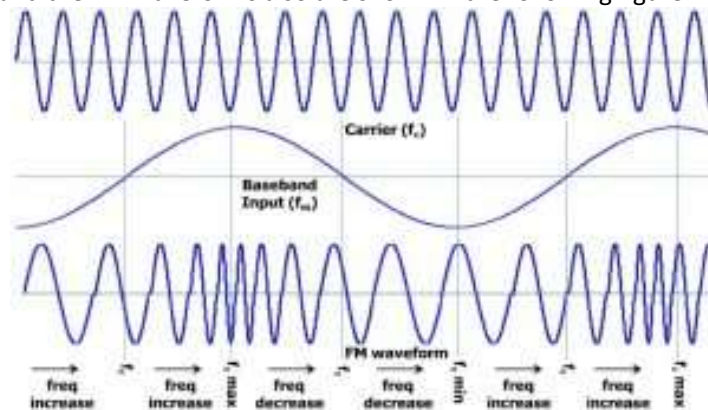
Lab 05 - Designing and Analyzing Frequency Modulator and Demodulation

1. Objective

The objective of this lab is to understand FM modulation & demodulation and gain a deeper understanding of its concept by analyzing it through simulating its behavior through Simulink

2. Introduction & Theoretical Background

The **frequency modulation** can be defined as; the frequency of the carrier signal is varied proportional to (in accordance with) the Amplitude of the input modulating signal. The input is a single tone sine wave. The carrier and the FM waveforms also are shown in the following figure.



The frequency of a carrier (f_c) will increase as the amplitude of modulating (input) signal increases. The carrier frequency will be maximum ($f_{c \max}$) when the input signal is at its peak. The carrier deviates maximum from its normal value. The frequency of a carrier will decrease as the amplitude of the modulating (input) signal decreases.

The frequency modulated signal is expressed as in time domain:

$$s(t) = A_c \cos\left[2\pi \int_{-\infty}^t \left(f_c + \frac{K_f}{2\pi} m(\lambda) d\lambda\right)\right]$$

Frequency Deviation

The amount of change in the carrier frequency produced, by the amplitude of the input modulating signal, is called **frequency deviation**. The Carrier frequency swings between $f_{\max}$ and $f_{\min}$ as the input varies in its amplitude. The difference between $f_{\max}$ and f_c is known as frequency deviation. $f_d = f_{\max} - f_c$. It is denoted by Δf

Similarly, the difference between f_c and $f_{\min}$ also is known as frequency deviation. $f_d = f_c - f_{\min}$

$$\Delta f = f_d = f_{\max} - f_c = f_c - f_{\min}$$

The **modulation index of FM** is defined as the ratio of the frequency deviation of the carrier to the frequency of the modulating signal

Modulation Index of FM = $\Delta f/f_m$

As a rule of thumb, often termed as Carson's Rule, 98% of the signal power in FM is contained within a bandwidth equal to the deviation frequency, plus the modulation frequency-doubled.

Carson's rule: Bandwidth of FM BWFM = $2 [\Delta f + f_m]$

Frequency Demodulation

a. PLL Demodulation

The PLL demodulates the FM signal using feedback force a Voltage-Controlled-Oscillator (VCO) to remain in phase with the carrier of the incoming signal. The message is recovered as the control input of the VCO . In the simulation experiment we used the VCO to demodulate the information signal to make life easier.

3. In-lab tasks

Task 1: Analyzing FM Modulated wave

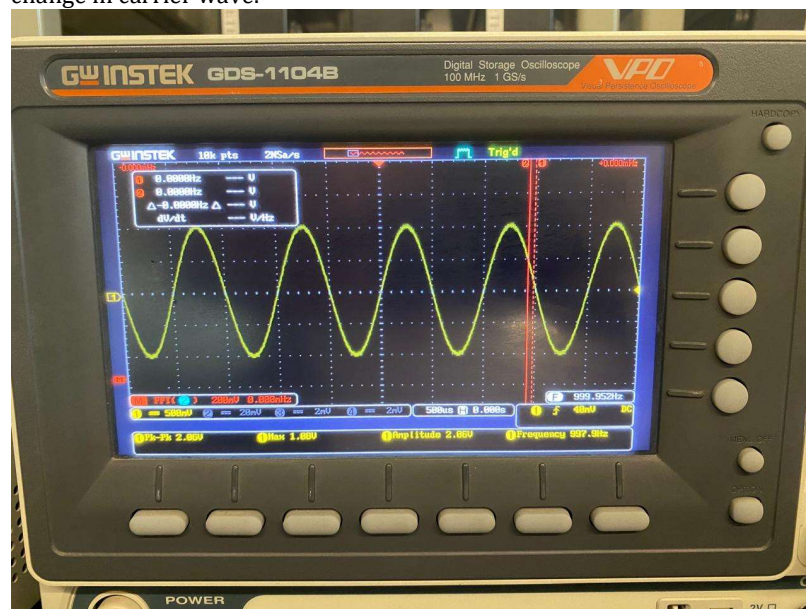
In this task we will analyze a FM modulated wave using function generator

1. Select a sine wave of frequency 1 KHz with 1 .00 Vpp from Waveform button in Function Generator
2. Select MOD→ FM
3. Choose following properties in FM Menu
 - Shape→ Sine wave
 - FM Frequency -120 Hz
 - Frequency Deviation = 0 Hz

Now Complete the following observation

1. Were you able to receive a FM modulated wave, with frequency deviation = 0 Hz

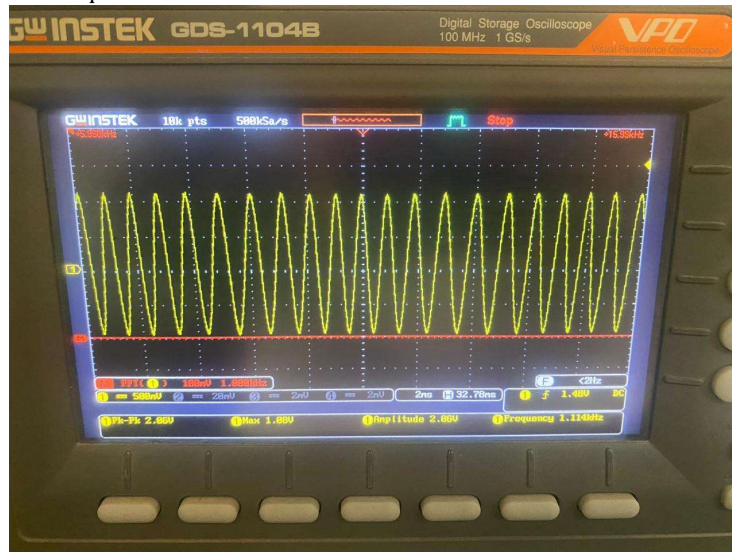
There is no FM wave when frequency deviation is set to zero. When we set frequency deviation to zero, the message signal causes no change in the frequency of the carrier, resulting in no change in carrier wave.



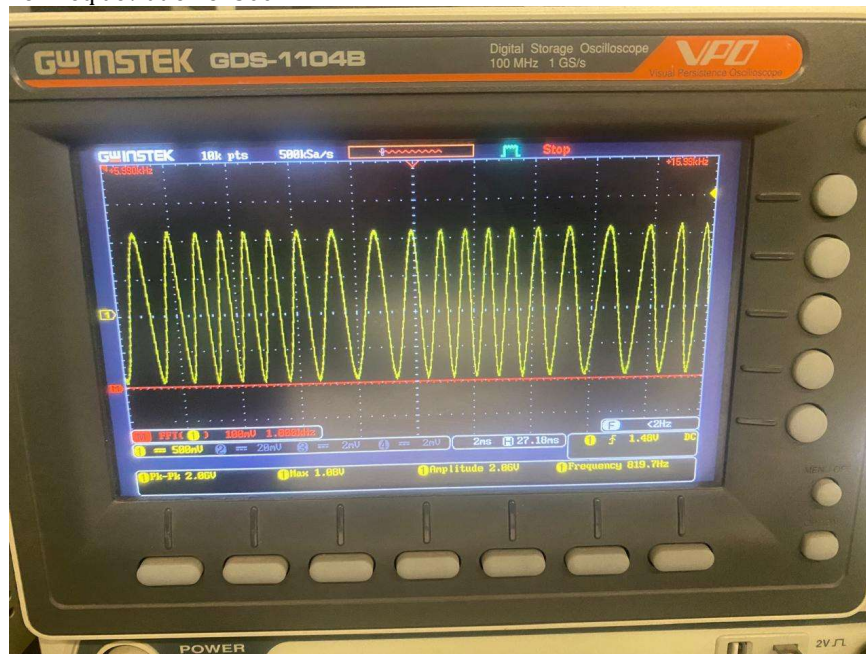
2. Were you able to receive a FM modulated wave, with frequency deviation = 120Hz and 300 Hz . Attach your screen shots

Yes, when setting frequency deviation is set to 120Hz or 300Hz, we do see a FM modulated wave.

For freq deviation of 120Hz:

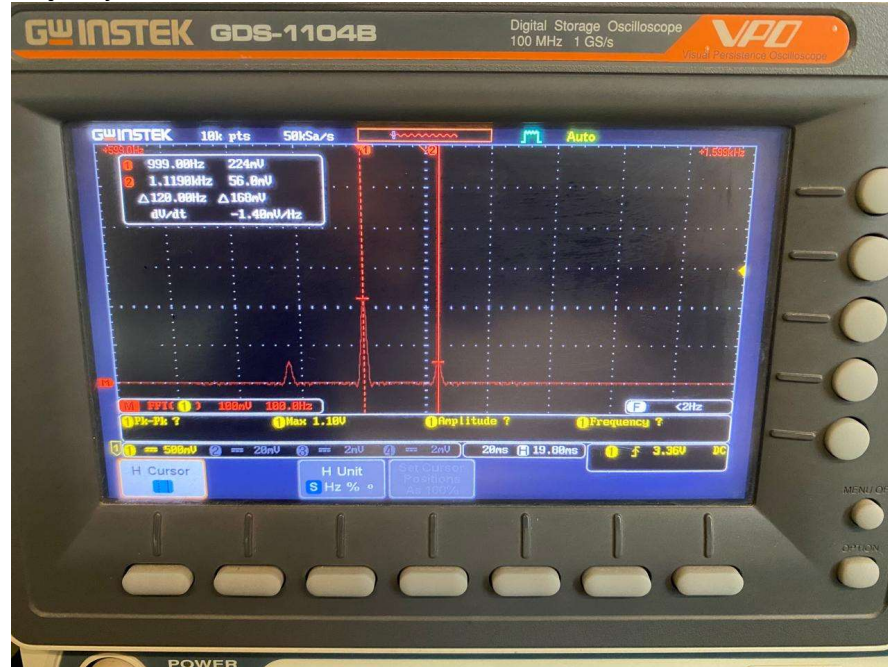


For freq deviation of 300Hz:



- Reset the frequency Deviation to 60 Hz . Analyze the above wave form on FFT domain. Can you create any logical link between sidebands and FM frequency? Attach the screenshot. How does increasing and decreasing the FM frequency effects the side bands.

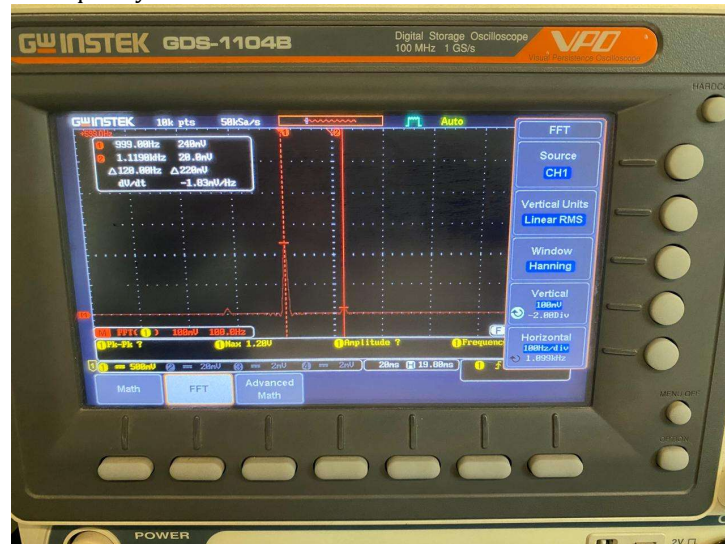
The center frequency in the FFT is the carrier band. Increasing the carrier wave frequency increases the distance between sidebands and the center carrier band, while decreasing the frequency reduces the distance.



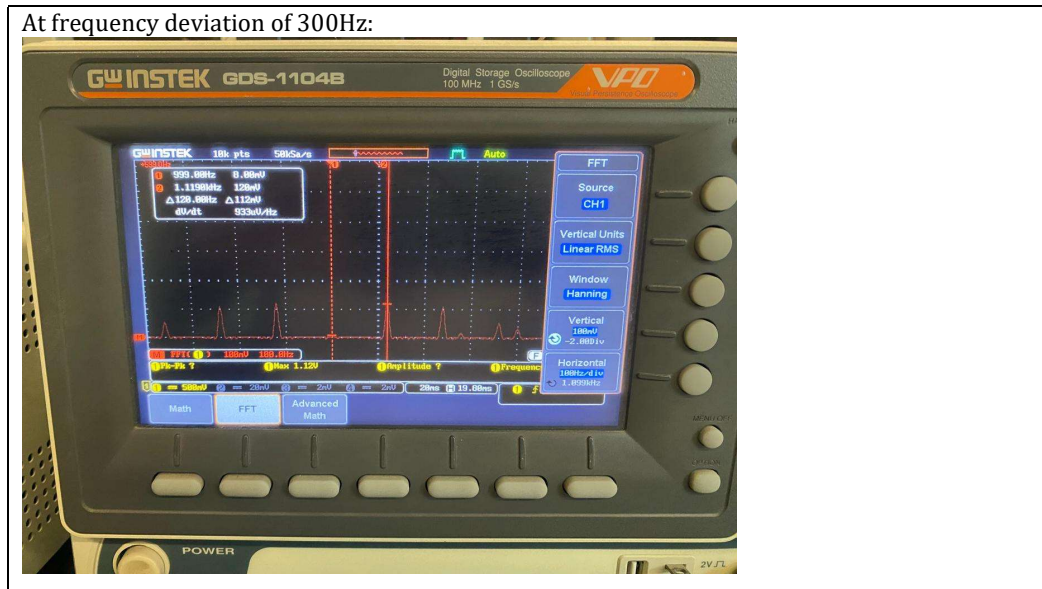
- does decreasing the FM deviation effects the side band and carrier signal, and how does increasing the FM deviation effects the side band and carrier signal. Record your observation below.

For low frequency deviation: Decreasing the frequency deviation compared to the modulating frequency reduces the amplitude of sidebands significantly.

At frequency deviation of 20Hz:



For high frequency deviation: Increasing the frequency deviation beyond the modulating frequency creates more sidebands of significant amplitude while decreasing the amplitude of the carrier band.



Task 2:

Simulate the following circuit and analyses its output.

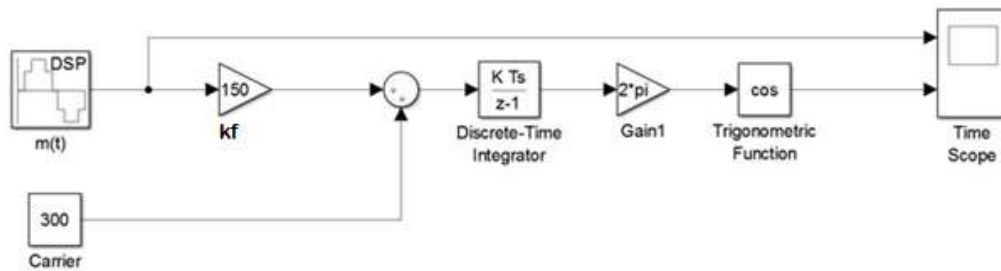


Figure 1 FM Modulator

The blocks' parameters are as described below:

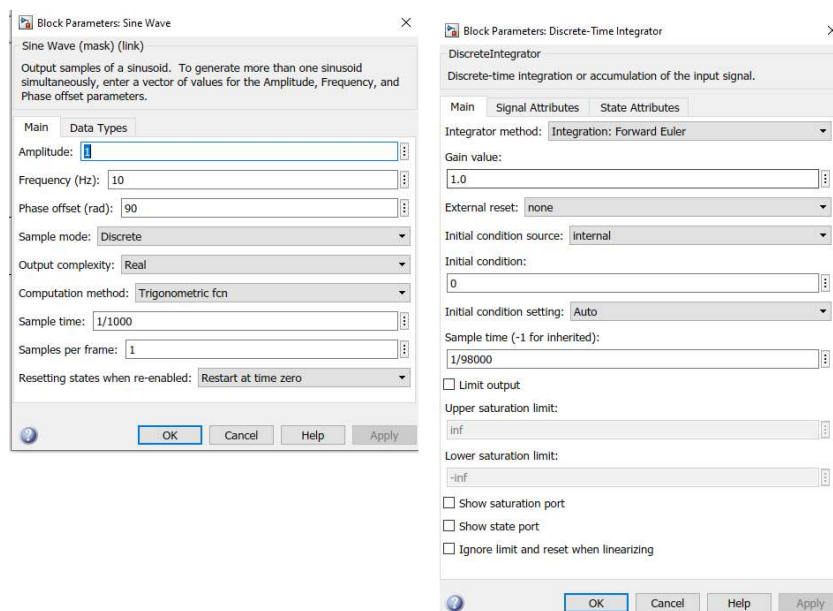
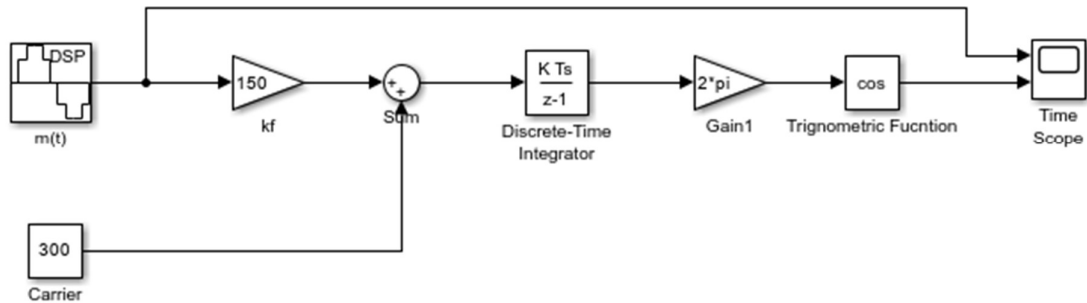


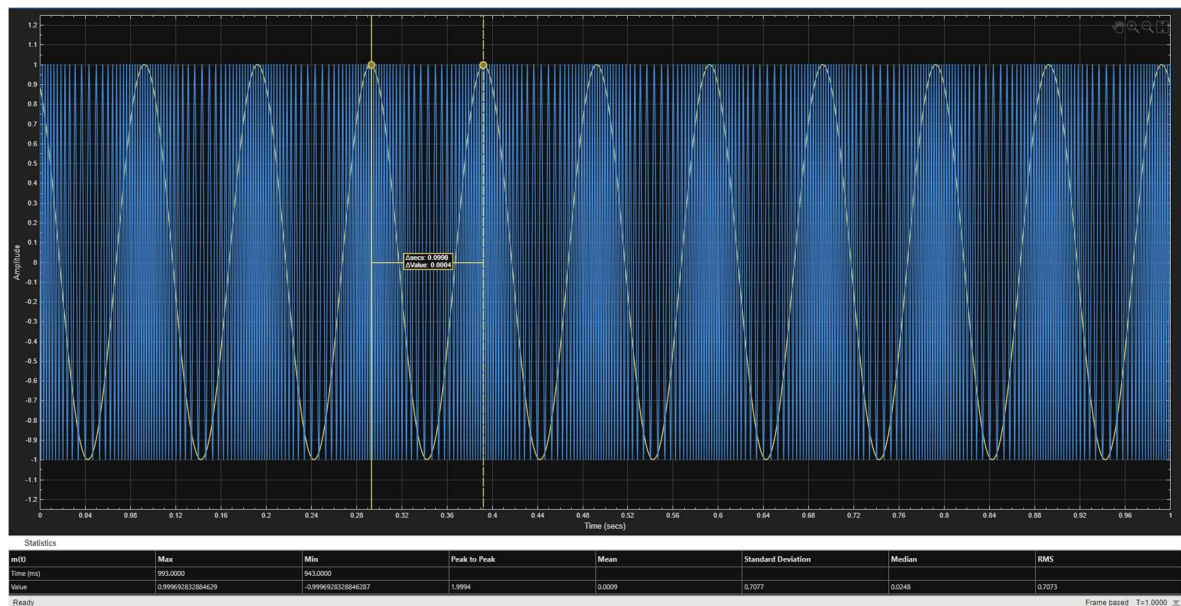
Figure 2 Block Parameter for FM Modulator

Attach the screenshot of labelled Simulink model, Output (Time scope) and comment on observed waveforms.

Simulink model:



Output on the time scope:



Observations: When the message signal's amplitude increases, the frequency of the modulated signal increases; similarly, when the amplitude of the message signal decreases, the frequency of the modulated signal decreases.

Task 3: Frequency Modulator and Demodulator

Design and implement a FM demodulator in Simulink without using the built-in FM Demodulator block.

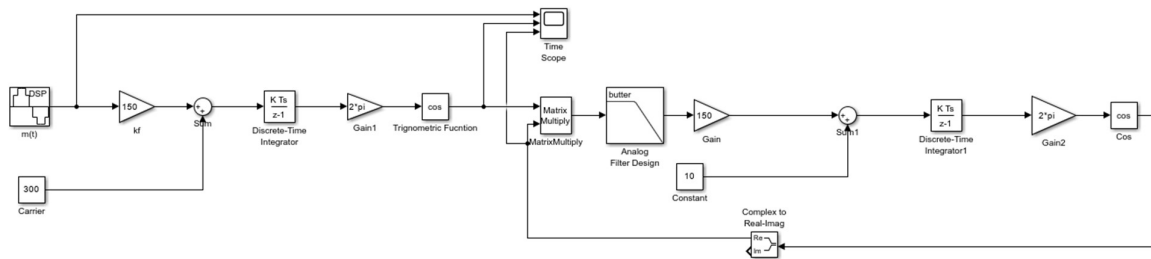
Attach the screenshot of your Simulink model, parameters setting and the plot of the demodulated output waveform.

We replicated the Continuous-Time VCO block through its equation taken from MATLAB website (<https://www.mathworks.com/help/comm/ref/continuoustimevco.html>) because the aforementioned block has been discontinued. To replicate it, we basically had to construct the equation using the basic blocks in Simulink.

$$y(t) = A_c \cos\left(2\pi f_c t + 2\pi k_c \int_0^t u(\tau) d\tau + \phi\right)$$

Continuous-Time VCO equation as per MATLAB website

Simulink model:



Parameter settings:

A_c = output amplitude = 1v as we want gain of one after cos block

ϕ = initial phase = 0 as the demodulated signals aligns well with the input signal so no need

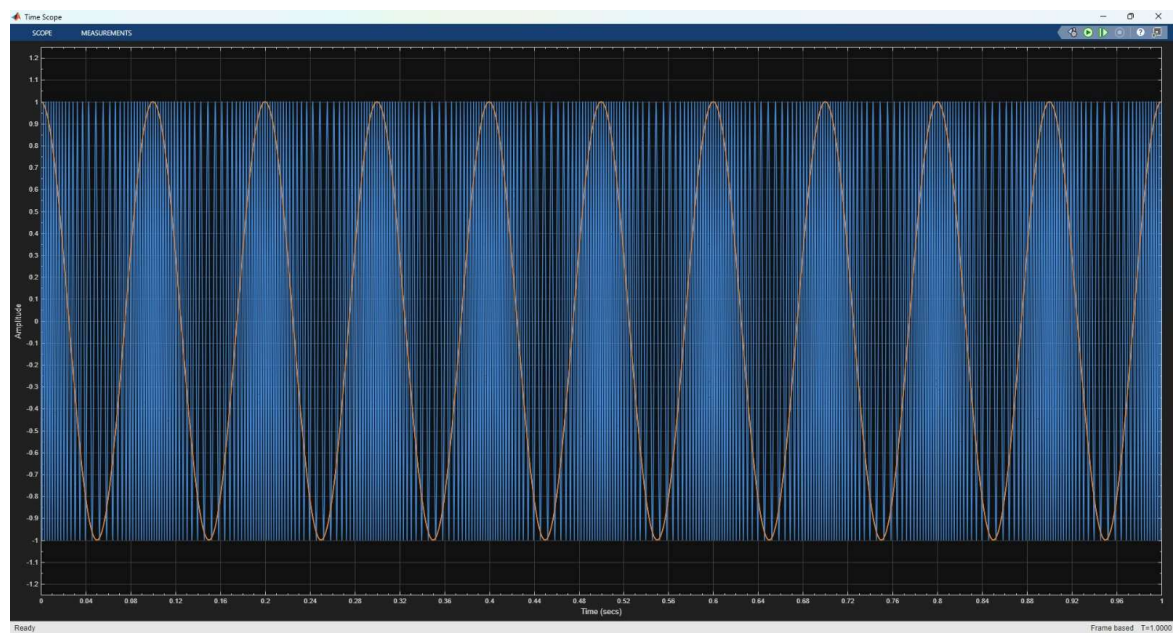
f_c = quiescent frequency = 10Hz as that is the input signal's frequency

k_c = input sensitivity = 50Hz/V (works best for our use case)

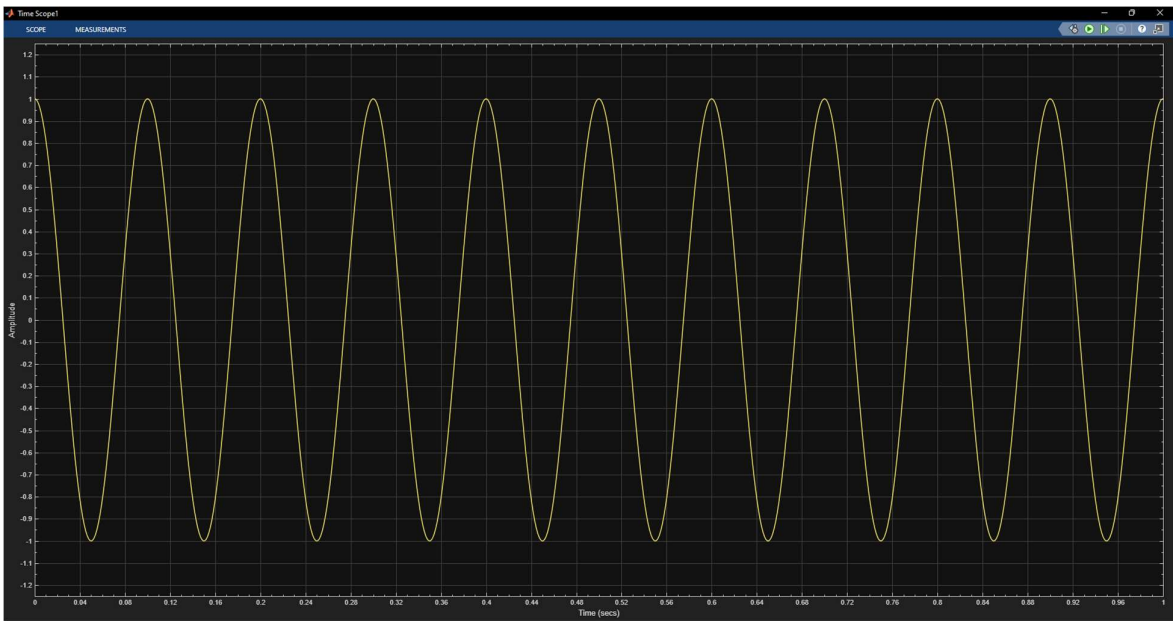
$u(\tau)$ = input signal from the filter = 2pi rad/s

The input to the integrator is $(f_c + k_c * u(\tau))$. After performing the integration and multiplying by 2π , it becomes $2\pi * (f_c * t + k_c * \text{integral}(u(\tau)))$. This is equivalent to the expression above, assuming $A_c = 1$ and $\phi = 0$.

Output plot:



Plot of the demodulated output waveform (isolated from the plot of all signals given above):



Frequency of recovered message signal	<u>Frequency:</u> 0.99Hz
Filter cutoff	<u>Filter cutoff frequency:</u> 30Hz

Name & ID: _____

Date: _____

EE-322L Analog and Digital Communications Lab

Fall 2025

Habib University

Dhanani School of Science & Engineering



Lab #5 Marks distribution:

		LR1=05	LR2=20	LR4=25	LR5=30	AR4=20
In-Lab Tasks	Task 1	/5		/5	/10	/20
	Task 2		/10	/10	/10	
	Task 3		/10	/10	/10	
Marks Obt.	/100					

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4< Level 2<=6)	(6< Level 3<=8)	(8< Level 4<=10)
LR1	Neat and Clean Circuit layout	Components are wired and didn't show neat and clean connections and minimal efforts shown.	Components are wired with untidy connection and didn't show neat and clean connections	Few but not all Components are wired with neat and clean connections	Complete components are wired with neat and clean tight connections. Given task completed in due time.
LR3	Troubleshooting	Unable to identify the fault/minimal effort shown	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and able to make necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included	Measurements are somewhat inaccurate and very imprecise. Observations are incomplete or recorded in a confusing way. Major errors using symbols, units and significant digits	Measurements are mostly accurate. Observations are generally complete. Minor errors using symbols, units and significant digits	Measurements are both accurate and precise. Observations are very thorough. Includes appropriate symbols, units and significant digits and completed in due time.
LR5	Results & Plots	Figures, graphs, tables contain errors or are poorly constructed, have missing titles, captions, units missing or incorrect, etc.	Most figures, graphs, tables OK, some still missing some important or required features	All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved	All figures, graphs, tables are correctly drawn and contain titles/captions.
AR4	*Report Submission	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

***Report:** Report will not be accepted after due date